

SRI LANKA INSTITUTE OF INFORMATION TECHNOLOGY

Faculty of Computing — Department of Software Engineering

Research Project — Periodic Log Entry Form

Project **R26-SE-009** | Agent 4 | **Malith P. Bandara** (IT22249166) | Reporting period **23.03.2026 – 25.08.2026**

1. PROJECT AND STUDENT DETAILS

Project ID	R26-SE-009
Research Topic	Self-Optimized LLM Multi-Agent System for Software Engineering
Supervisor	Prof. Nuwan Kodagoda
Co-Supervisor	Mr. Eishan Weerasinghe
Student Name	Malith P. Bandara
Student Registration No.	IT22249166
Individual Component	Agent 4 — Deployment Engineer (AI DevOps / Deployment) and the DevOps user interface
Reporting Period	23 March 2026 to 25 August 2026
Phase Covered	Requirement, implementation and integration phase, representing approximately 65% of the planned project work.

2. SUMMARY OF RP WORK CARRIED OUT DURING THE REPORTING PERIOD

During this reporting period Agent 4 was developed as the deployment and DevOps component of the multi-agent system, together with the DevOps-facing user interface. The early part of the period established deployment artefact generation, run tracking, readiness reporting and the structure required for Docker, GitHub Actions, AWS-style deployment evidence and release packaging, alongside two iterations of the deployment interface covering workflow diffing, timeline replay and artefact presentation. The middle of the period contributed shared studio screens — the design selector, the builder timeline and the generated-page preview. The later part of the period delivered the CI quality and security gates — SonarCloud analysis and Snyk dependency and infrastructure scanning — the functional integration dashboard and diagnostics, and the version 2 deployment package with hardened credential handling, environment and database validation, deployment event tracking and provider-backed release support.

3. DATE-WISE RP WORK (43 ENTRIES, VERIFIED MONTHLY)

MARCH 2026			5 ENTRIES
NO.	DATE	RP WORK CARRIED OUT	
1	23.03.2026	Defined the scope of Agent 4 as the AI Deployment / DevOps Engineer responsible for packaging, deployment readiness and release support after QA approval.	
2	24.03.2026	Reviewed the expected deployment outputs for the project, including Docker assets, compose files, workflow pipelines, cloud configuration and release evidence.	
3	26.03.2026	Designed the Agent 4 workflow for moving from approved source code to deployment artefacts and readiness reporting.	
4	28.03.2026	Planned support for Docker, GitHub Actions, AWS ECS/ECR configuration, environment templates and API validation evidence.	
5	31.03.2026	Identified the pipeline checkpoints needed for source validation, service discovery, security review, packaging, deployment and documentation.	
MONTHLY VERIFICATION — MARCH 2026			
SUPERVISOR'S COMMENTS FOR MARCH 2026		SUPERVISOR'S SIGNATURE	CO-SUPERVISOR'S SIGNATURE
		Signature & Date	Signature & Date

APRIL 2026		15 ENTRIES
NO.	DATE	RP WORK CARRIED OUT
6	02.04.2026	Defined the artefact set required from Agent 4, including compose files, workflow files, task definitions, summaries and evidence bundles.
7	04.04.2026	Refined readiness scoring and deployment-gate logic so unresolved QA issues could block release preparation in a controlled way.
8	07.04.2026	Coordinated the deployment handoff with the group workflow so Agent 4 depends on Agent 3 pass status and produces visible artefacts for the shared demo.
9	09.04.2026	Planned the reporting structure for operational evidence such as ECS services, ECR images, logs, API validation and checklist-driven documentation.
10	18.04.2026	Implemented the animated conversation area for the deployment assistant interface.
11	18.04.2026	Resolved the reported repository security findings and cleared the remaining scan issues.
12	19.04.2026	Added workflow change viewing and filtering so modified pipeline files can be inspected before a release.
13	19.04.2026	Added timeline replay for deployment runs and implemented the smart diff preview section.
14	19.04.2026	Improved the deployment artefact area and the way generated files are presented.
15	20.04.2026	Removed superseded interface sections and upgraded the deployment interface presentation.
16	21.04.2026	Completed version 2 of the deployment interface after removing the remaining unnecessary sections.
17	23.04.2026	Implemented the Agent 4 backend in code, including deploy start/status endpoints, run tracking and artefact generation for docker-compose, GitHub workflow and deployment summary files.
18	24.04.2026	Established the shared DevOps Center page structure and updated the conversation section to handle missing intake questions.
19	27.04.2026	Completed version 2.1 of the deployment workspace, expanding services, workflows, ECS, ECR, CloudWatch, readiness scores and evidence checklist items.
20	28.04.2026	Consolidated the deployment-phase outcomes and prepared the supervisor-ready summary for the reporting period.

MONTHLY VERIFICATION — APRIL 2026		
SUPERVISOR'S COMMENTS FOR APRIL 2026	SUPERVISOR'S SIGNATURE	CO-SUPERVISOR'S SIGNATURE
	Signature & Date	Signature & Date

MAY 2026

3 ENTRIES

NO.	DATE	RP WORK CARRIED OUT
21	05.05.2026	Built the design-selector screens and the preset browsing flow in the shared workspace.
22	06.05.2026	Implemented the builder timeline and progress states shown while generation is running.
23	08.05.2026	Added the generated-page preview and the coder pane used to inspect produced files.

MONTHLY VERIFICATION — MAY 2026		
SUPERVISOR'S COMMENTS FOR MAY 2026	SUPERVISOR'S SIGNATURE	CO-SUPERVISOR'S SIGNATURE
	Signature & Date	Signature & Date

JUNE 2026

3 ENTRIES

NO.	DATE	RP WORK CARRIED OUT
24	10.06.2026	Reviewed the deployment path end to end and listed the checks required before a release can be started.
25	17.06.2026	Compared container, CI and cloud provider options and selected the target deployment routes for version 2.
26	24.06.2026	Planned the CI quality and security gates to be added to the repository workflows.

MONTHLY VERIFICATION — JUNE 2026		
SUPERVISOR'S COMMENTS FOR JUNE 2026	SUPERVISOR'S SIGNATURE	CO-SUPERVISOR'S SIGNATURE
	Signature & Date	Signature & Date

JULY 2026			9 ENTRIES		
NO.	DATE	RP WORK CARRIED OUT			
27	23.07.2026	Added the SonarCloud quality-analysis workflow and configured project analysis for the repository.			
28	23.07.2026	Added the Snyk dependency and infrastructure-as-code security workflow.			
29	23.07.2026	Built the functional integration access dashboard and the integration status service.			
30	23.07.2026	Enabled the functional integration controls in the DevOps Center and exposed connection diagnostics.			
31	23.07.2026	Added the GitHub CLI integration card and documented the SonarCloud and Snyk connection steps.			
32	23.07.2026	Documented the integration environment configuration and cleaned generated files out of version control.			
33	24.07.2026	Fixed the CI tests for the SonarCloud and Snyk workflows and corrected the Docker validator test environment.			
34	24.07.2026	Fixed the front-end build and upgraded Next.js and Sharp to clear the reported security advisories.			
35	24.07.2026	Adjusted the pipeline to skip the unavailable Snyk Code and IaC scans so results are reported accurately.			
MONTHLY VERIFICATION — JULY 2026					
SUPERVISOR'S COMMENTS FOR JULY 2026		SUPERVISOR'S SIGNATURE		CO-SUPERVISOR'S SIGNATURE	
		Signature & Date		Signature & Date	

AUGUST 2026			8 ENTRIES	
NO.	DATE	RP WORK CARRIED OUT		
36	21.08.2026	Refreshed the deployment package foundation and reorganised it into the modular deploy_agent structure.		
37	22.08.2026	Hardened credential and secret handling for deployment operations.		
38	22.08.2026	Cleaned the repository and updated the component documentation.		
39	23.08.2026	Added deployment event and state tracking for running releases.		
40	23.08.2026	Added environment validation and MongoDB connectivity checks carried out before a deployment starts.		
41	24.08.2026	Refreshed the deployment package and the flow contracts shared with the QA stage.		
42	24.08.2026	Strengthened validation and the deployment tool interfaces.		
43	25.08.2026	Added Vercel authentication and API support for provider-backed deployment.		
MONTHLY VERIFICATION — AUGUST 2026				
SUPERVISOR'S COMMENTS FOR AUGUST 2026		SUPERVISOR'S SIGNATURE		CO-SUPERVISOR'S SIGNATURE
		Signature & Date		Signature & Date
4. CONSOLIDATED MONTHLY VERIFICATION				
MONTH	ENTRIES LOGGED	SUPERVISOR'S SIGNATURE		DATE
March 2026	5			
April 2026	15			
May 2026	3			
June 2026	3			
July 2026	9			
August 2026	8			
Total	43			

4. CONSOLIDATED MONTHLY VERIFICATION

MONTH	ENTRIES LOGGED	SUPERVISOR'S SIGNATURE	DATE
March 2026	5		
April 2026	15		
May 2026	3		
June 2026	3		
July 2026	9		
August 2026	8		
Total	43		

5. PROJECT WORK PLANNED BEFORE THE NEXT EVALUATION PERIOD

- Complete provider-backed deployment through Vercel and the AWS route, including release generation and rollback handling.
- Finish deployment monitoring — pipeline, repository, security and evidence views — in the DevOps Center.
- Strengthen readiness scoring and secret handling so releases are gated on verifiable QA acceptance.
- Extend the generated deployment artefacts into more complete service-specific container and cloud configurations.

5. Tighten the live handoff from Agent 3 to Agent 4 so deployment starts only after a strong QA pass signal.

6. DECLARATION AND SIGNATURES

I certify that the work recorded in this log book for the period 23 March 2026 to 25 August 2026 was carried out by me as the individual component contribution to research project R26-SE-009, and that the group-level contributions recorded here were made jointly with the other project members.

STUDENT _____ Malith P. Bandara (IT22249166) — Signature & Date	SUPERVISOR _____ Prof. Nuwan Kodagoda — Signature & Date
CO-SUPERVISOR _____ Mr. Eishan Weerasinghe — Signature & Date	OVERALL SUPERVISOR COMMENTS _____ Comments